

MA2040 Jan-Apr 2024
Problem Sheet 3

1. Define a random variable and explain its role in probability theory. Provide examples of both discrete and continuous random variables.
2. Let X represent the number of heads obtained when flipping three fair coins. Determine the probability mass function (PMF) of X and calculate the expected value and variance of X .
3. Consider a discrete random variable Y with the following PMF:

$$P(Y = k) = \frac{1}{2^k}, \quad \text{for } k = 1, 2, 3, \dots$$

Compute the expected value and variance of Y .

4. Define a binomial random variable Z with parameters $n = 5$ and $p = 0.4$. Determine the probability mass function of Z and find $P(Z \leq 2)$.
5. Suppose X follows a Poisson distribution with parameter $\lambda = 3$. Calculate $P(X = 2)$ and $P(X > 5)$.
6. Let Y be a continuous random variable representing the time (in hours) it takes to complete a task, with probability density function given by:

$$f(y) = \begin{cases} 2y & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the cumulative distribution function (CDF) of Y and determine $P(Y > 0.5)$.

7. Define a normal random variable Z with mean $\mu = 10$ and standard deviation $\sigma = 2$. Compute $P(8 < Z < 12)$.

8. Consider a continuous random variable X with probability density function:

$$f(x) = \begin{cases} cx^2 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of c that makes $f(x)$ a valid probability density function.

9. Define a geometric random variable Y representing the number of trials needed to achieve the first success, with success probability $p = 0.3$. Calculate $P(Y = 3)$ and $P(Y \leq 4)$.